Machine Learning Fundamentals

Machine learning is a field of artificial intelligence in which systems learn patterns from examples rather than relying only on hand-written rules. A model is trained using data and then used to make predictions on new inputs.

The three common learning settings are supervised learning, unsupervised learning, and reinforcement learning. Supervised learning uses labelled examples. Unsupervised learning looks for structure in unlabelled data. Reinforcement learning learns actions through rewards and penalties.

Supervised Learning and Evaluation

In supervised learning, classification predicts a category, such as spam or not spam. Regression predicts a numeric value, such as a house price.

A dataset is often divided into training, validation, and test sets. The training set is used to fit model parameters. The validation set supports model selection and tuning. The test set estimates performance on held-out examples and should not be repeatedly used to tune the model.

Accuracy is the fraction of predictions that are correct. Precision is true positives divided by predicted positives. Recall is true positives divided by actual positives. The F1 score is the harmonic mean of precision and recall.

Overfitting and Generalisation

Overfitting occurs when a model fits details or noise in its training data and performs poorly on new examples. Generalisation is the ability to perform well on unseen data.

Common ways to reduce overfitting include collecting more representative data, regularisation, simplifying the model, early stopping, and using a validation set. No single technique guarantees good generalisation; evaluation should reflect the intended use.